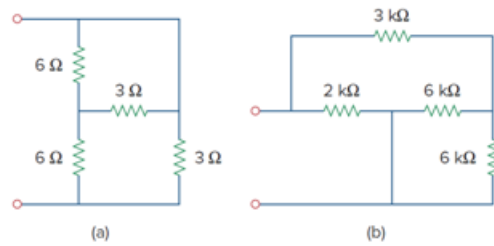


Problem

Evaluate R_{eq} looking into each set of terminals for each of the circuits shown in Fig. 2.103.

Figure 2.103:



Step-by-step solution

Step 1 of 5

(a)

Consider the circuit diagram of textbook Figure 2.103(a). The upper 6Ω and 3Ω resistors are in parallel. Therefore their equivalent resistance is,

$$(6 \parallel 3) = \frac{(6)(3)}{9} \\ = 2\Omega$$

The resultant circuit is as shown in Figure 1.

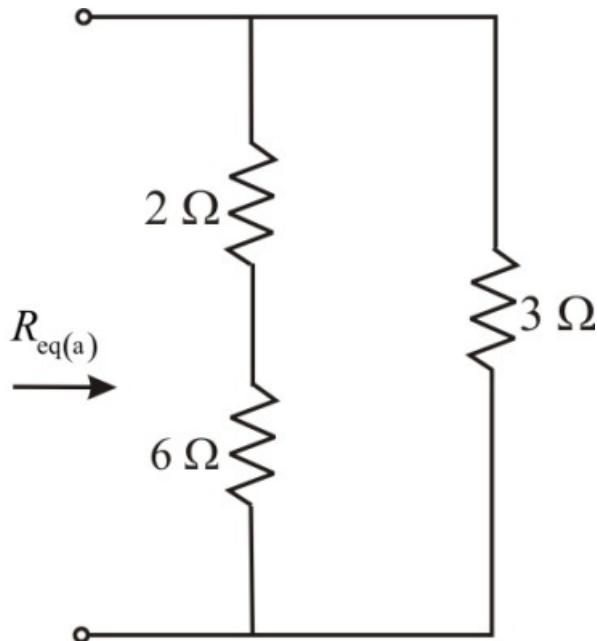


Figure 1

Step 2 of 5

The equivalent resistance of Figure 1 is,

$$\begin{aligned} R_{eq(a)} &= (2 + 6) \parallel 3 \\ &= \frac{(8)(3)}{11} \\ &= 2.18 \Omega \end{aligned}$$

Therefore the equivalent resistance of the textbook Figure 2.103(a) is 2.18 Ω .

Step 3 of 5

(b)

Consider the circuit diagram of textbook Figure 2.103(b). The lower $6 \text{ k}\Omega$ and $6 \text{ k}\Omega$ resistors are in parallel. Therefore their equivalent resistance (R_{eq}) is,

$$\begin{aligned} R_{eq} &= (6 \text{ k} \parallel 6 \text{ k}) \\ &= \frac{(6 \text{ k})(6 \text{ k})}{12 \text{ k}} \\ &= 3 \text{ k}\Omega \end{aligned}$$

The resultant circuit is as shown in Figure 2.

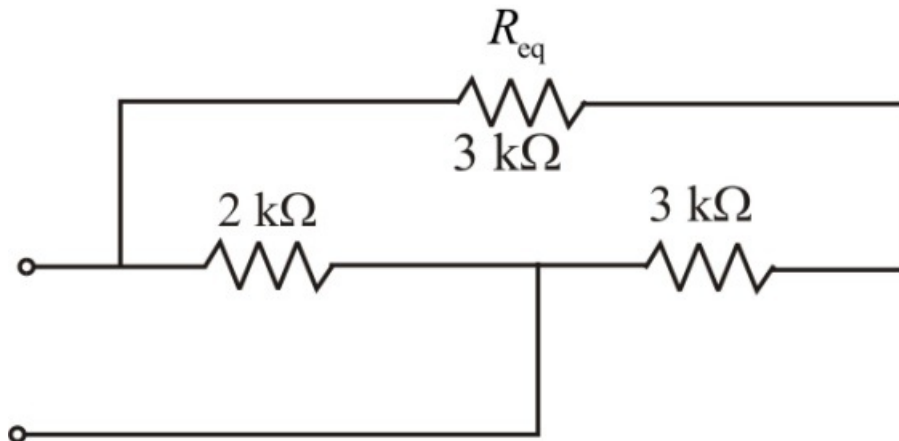


Figure 2

Step 4 of 5

From Figure 2, the two $3 \text{ k}\Omega$ resistors are in series. Therefore the resultant circuit diagram is as shown in Figure 3.

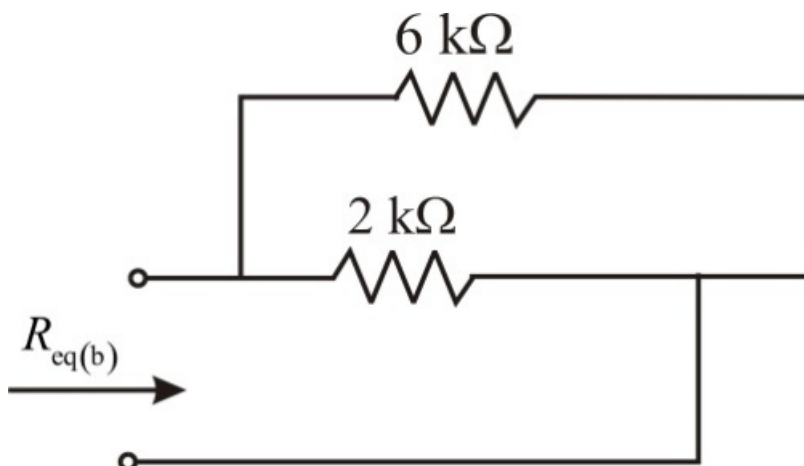


Figure 3

Step 5 of 5

From Figure 3, the equivalent resistance is,

$$\begin{aligned} R_{\text{eq(b)}} &= (6\text{k}) \parallel (2\text{k}) \\ &= \frac{(6\text{k})(2\text{k})}{8\text{k}} \\ &= 1.5 \text{ k}\Omega \end{aligned}$$

Therefore, the equivalent resistance of the textbook Figure 2.103(b) is 1.5 k Ω .